6_Databases_Data_Science_Assignment_Introduction

January 30, 2024

Assignment: Notebook for Peer Assignment

1 Introduction

Using this Python notebook you will:

1. Understand three Chicago datasets
2. Load the three datasets into three tables in a SQLIte database
3. Execute SQL queries to answer assignment questions

1.1 Understand the datasets

To complete the assignment problems in this notebook you will be using three datasets that are available on the city of Chicago's Data Portal:

1. Socioeconomic Indicators in Chicago
2. Chicago Public Schools
3. Chicago Crime Data

1.1.1 1. Socioeconomic Indicators in Chicago

This dataset contains a selection of six socioeconomic indicators of public health significance and a "hardship index," for each Chicago community area, for the years 2008 – 2012.

A detailed description of this dataset and the original dataset can be obtained from the Chicago Data Portal at: https://data.cityofchicago.org/Health-Human-Services/Census-Data-Selected-socioeconomic-indicators-in-C/kn9c-c2s2

1.1.2 2. Chicago Public Schools

This dataset shows all school level performance data used to create CPS School Report Cards for the 2011-2012 school year. This dataset is provided by the city of Chicago's Data Portal.

A detailed description of this dataset and the original dataset can be obtained from the Chicago Data Portal at: https://data.cityofchicago.org/Education/Chicago-Public-Schools-Progress-Report-Cards-2011-/9xs2-f89t

1.1.3 3. Chicago Crime Data

This dataset reflects reported incidents of crime (with the exception of murders where data exists for each victim) that occurred in the City of Chicago from 2001 to present, minus the most recent seven days.

A detailed description of this dataset and the original dataset can be obtained from the Chicago Data Portal at: https://data.cityofchicago.org/Public-Safety/Crimes-2001-to-present/ijzp-q8t2

1.1.4 Download the datasets

This assignment requires you to have these three tables populated with a subset of the whole datasets.

In many cases the dataset to be analyzed is available as a .CSV (comma separated values) file, perhaps on the internet. Click on the links below to download and save the datasets (.CSV files):

- Chicago Census Data
- Chicago Public Schools
- Chicago Crime Data

NOTE: Ensure you have downloaded the datasets using the links above instead of directly from the Chicago Data Portal. The versions linked here are subsets of the original datasets and have some of the column names modified to be more database friendly which will make it easier to complete this assignment.

1.1.5 Store the datasets in database tables

To analyze the data using SQL, it first needs to be loaded into SQLite DB. We will create three tables in as under:

1. **CENSUS_DATA**
2. **CHICAGO_PUBLIC_SCHOOLS**
3. **CHICAGO_CRIME_DATA**

Let us now load the ipython-sql extension and establish a connection with the database

- Here you will be loading the csv files into the pandas Dataframe and then loading the data into the above mentioned sqlite tables.
- Next you will be connecting to the sqlite database **FinalDB**.

Refer to the previous lab for hints .

Hands-on Lab: Analyzing a real World Data Set

[1]:
```
import csv, sqlite3
con = sqlite3.connect('FinalDB.db')
cur = con.cursor()
```

[2]:
```
%load_ext sql
```

[3]:
```
%sql sqlite:///FinalDB.db
```

[3]:
```
'Connected: @FinalDB.db'
```

[4]:
```
!pip install -q pandas==1.1.5
```

[5]:
```
import pandas as pd
df_census = pd.read_csv("https://cf-courses-data.s3.us.cloud-object-storage.
↪appdomain.cloud/IBMDeveloperSkillsNetwork-DB0201EN-SkillsNetwork/labs/
↪FinalModule_Coursera_V5/data/ChicagoCensusData.csv", header = 0)
df_census.to_sql('CENSUS_DATA', con, if_exists = 'replace', index = False,␣
↪method = 'multi')
df_schools = pd.read_csv("https://cf-courses-data.s3.us.cloud-object-storage.
↪appdomain.cloud/IBMDeveloperSkillsNetwork-DB0201EN-SkillsNetwork/labs/
↪FinalModule_Coursera_V5/data/ChicagoPublicSchools.csv", header = 0)
df_schools.to_sql('CHICAGO_PUBLIC_SCHOOLS_DATA', con, if_exists = 'replace',␣
↪index = False, method = 'multi')
df_crime = pd.read_csv("https://cf-courses-data.s3.us.cloud-object-storage.
↪appdomain.cloud/IBMDeveloperSkillsNetwork-DB0201EN-SkillsNetwork/labs/
↪FinalModule_Coursera_V5/data/ChicagoCrimeData.csv", header = 0)
df_crime.to_sql('CHICAGO_CRIME_DATA', con, if_exists = 'replace', index =␣
↪False, method = 'multi')
```

```
/home/jupyterlab/conda/envs/python/lib/python3.7/site-
packages/pandas/core/generic.py:2615: UserWarning: The spaces in these column
names will not be changed. In pandas versions < 0.14, spaces were converted to
underscores.
  method=method,
```

[6]:
```
df_census.head()
```

[6]:
```
   COMMUNITY_AREA_NUMBER COMMUNITY_AREA_NAME  PERCENT_OF_HOUSING_CROWDED  \
0                    1.0         Rogers Park                         7.7
1                    2.0         West Ridge                          7.8
2                    3.0             Uptown                          3.8
3                    4.0     Lincoln Square                          3.4
4                    5.0       North Center                          0.3

   PERCENT_HOUSEHOLDS_BELOW_POVERTY  PERCENT_AGED_16__UNEMPLOYED  \
0                              23.6                          8.7
1                              17.2                          8.8
2                              24.0                          8.9
3                              10.9                          8.2
4                               7.5                          5.2

   PERCENT_AGED_25__WITHOUT_HIGH_SCHOOL_DIPLOMA  \
0                                          18.2
1                                          20.8
2                                          11.8
3                                          13.4
4                                           4.5
```

```
   PERCENT_AGED_UNDER_18_OR_OVER_64  PER_CAPITA_INCOME  HARDSHIP_INDEX
0                              27.5              23939            39.0
1                              38.5              23040            46.0
2                              22.2              35787            20.0
3                              25.5              37524            17.0
4                              26.2              57123             6.0
```

[7]:
```
df_schools.head()
```

[7]:
```
   School_ID                                     NAME_OF_SCHOOL  \
0     610038                   Abraham Lincoln Elementary School
1     610281  Adam Clayton Powell Paideia Community Academy …
2     610185                 Adlai E Stevenson Elementary School
3     609993                      Agustin Lara Elementary Academy
4     610513                        Air Force Academy High School

  Elementary, Middle, or High School         Street_Address     City State  \
0                                 ES         615 W Kemper Pl  Chicago    IL
1                                 ES  7511 S South Shore Dr  Chicago    IL
2                                 ES      8010 S Kostner Ave  Chicago    IL
3                                 ES      4619 S Wolcott Ave  Chicago    IL
4                                 HS        3630 S Wells St  Chicago    IL

   ZIP_Code    Phone_Number  \
0     60614  (773) 534-5720
1     60649  (773) 535-6650
2     60652  (773) 535-2280
3     60609  (773) 535-4389
4     60609  (773) 535-1590

                                                Link  \
0  http://schoolreports.cps.edu/SchoolProgressRep…
1  http://schoolreports.cps.edu/SchoolProgressRep…
2  http://schoolreports.cps.edu/SchoolProgressRep…
3  http://schoolreports.cps.edu/SchoolProgressRep…
4  http://schoolreports.cps.edu/SchoolProgressRep…

                  Network_Manager  … Freshman_on_Track_Rate__  \
0       Fullerton Elementary Network  …                      NDA
1          Skyway Elementary Network  …                      NDA
2          Midway Elementary Network  …                      NDA
3        Pershing Elementary Network  …                      NDA
4  Southwest Side High School Network  …                     91.8

  X_COORDINATE Y_COORDINATE   Latitude  Longitude COMMUNITY_AREA_NUMBER  \
0  1171699.458  1915829.428  41.924497 -87.644522                     7
```

```
1  1196129.985  1856209.466  41.760324 -87.556736                     43
2  1148427.165  1851012.215  41.747111 -87.731702                     70
3  1164504.290  1873959.199  41.809757 -87.672145                     61
4  1175177.622  1880745.126  41.828146 -87.632794                     34

  COMMUNITY_AREA_NAME  Ward Police_District                     Location
0        LINCOLN PARK    43              18  (41.92449696, -87.64452163)
1         SOUTH SHORE     7               4  (41.76032435, -87.55673627)
2            ASHBURN    13               8  (41.74711093, -87.73170248)
3           NEW CITY    20               9    (41.8097569, -87.6721446)
4       ARMOUR SQUARE    11               9  (41.82814609, -87.63279369)

[5 rows x 78 columns]
```

[8]:
```
df_crime.head()
```

[8]:
```
         ID CASE_NUMBER        DATE                   BLOCK IUCR  \
0   3512276    HK587712  2004-08-28        047XX S KEDZIE AVE  890
1   3406613    HK456306  2004-06-26  009XX N CENTRAL PARK AVE  820
2   8002131    HT233595  2011-04-04        043XX S WABASH AVE  820
3   7903289    HT133522  2010-12-30      083XX S KINGSTON AVE  840
4  10402076    HZ138551  2016-02-02           033XX W 66TH ST  820

  PRIMARY_TYPE                 DESCRIPTION          LOCATION_DESCRIPTION  \
0        THEFT               FROM BUILDING             SMALL RETAIL STORE
1        THEFT              $500 AND UNDER                          OTHER
2        THEFT              $500 AND UNDER  NURSING HOME/RETIREMENT HOME
3        THEFT  FINANCIAL ID THEFT: OVER $300                  RESIDENCE
4        THEFT              $500 AND UNDER                          ALLEY

   ARREST  DOMESTIC  …  DISTRICT  WARD  COMMUNITY_AREA_NUMBER  FBICODE  \
0   False     False  …         9  14.0                   58.0        6
1   False     False  …        11  27.0                   23.0        6
2   False     False  …         2   3.0                   38.0        6
3   False     False  …         4   7.0                   46.0        6
4   False     False  …         8  15.0                   66.0        6

  X_COORDINATE  Y_COORDINATE  YEAR   LATITUDE  LONGITUDE  \
0    1155838.0     1873050.0  2004  41.807441 -87.703956
1    1152206.0     1906127.0  2004  41.898280 -87.716406
2    1177436.0     1876313.0  2011  41.815933 -87.624642
3    1194622.0     1850125.0  2010  41.743665 -87.562463
4    1155240.0     1860661.0  2016  41.773455 -87.706480

                      LOCATION
0    (41.8074405, -87.703955849)
1  (41.898279962, -87.716405505)
```

```
2  (41.815933131, -87.624642127)
3  (41.743665322, -87.562462756)
4  (41.773455295, -87.706480471)

[5 rows x 21 columns]
```

[9]:
```
# Retrieving the list of all tables in the database
%sql SELECT name FROM sqlite_master WHERE type = 'table';
```

```
 * sqlite:///FinalDB.db
Done.
```

[9]:
```
[('CENSUS_DATA',), ('CHICAGO_PUBLIC_SCHOOLS_DATA',), ('CHICAGO_CRIME_DATA',)]
```

[10]:
```
# Retrieving the number of columns in the CENSUS_DATA table
%sql SELECT COUNT(NAME) FROM PRAGMA_TABLE_INFO('CENSUS_DATA');
```

```
 * sqlite:///FinalDB.db
Done.
```

[10]:
```
[(9,)]
```

[11]:
```
# Retrieving all column names in the CENSUS_DATA table along with their␣
 ↪datatypes and length
%sql SELECT name, type, length(type) FROM PRAGMA_TABLE_INFO('CENSUS_DATA');
```

```
 * sqlite:///FinalDB.db
Done.
```

[11]:
```
[('COMMUNITY_AREA_NUMBER', 'REAL', 4),
 ('COMMUNITY_AREA_NAME', 'TEXT', 4),
 ('PERCENT_OF_HOUSING_CROWDED', 'REAL', 4),
 ('PERCENT_HOUSEHOLDS_BELOW_POVERTY', 'REAL', 4),
 ('PERCENT_AGED_16__UNEMPLOYED', 'REAL', 4),
 ('PERCENT_AGED_25__WITHOUT_HIGH_SCHOOL_DIPLOMA', 'REAL', 4),
 ('PERCENT_AGED_UNDER_18_OR_OVER_64', 'REAL', 4),
 ('PER_CAPITA_INCOME', 'INTEGER', 7),
 ('HARDSHIP_INDEX', 'REAL', 4)]
```

[12]:
```
# Retrieving the number of columns in the CHICAGO_PUBLIC_SCHOOLS_DATA table
%sql SELECT COUNT(NAME) FROM PRAGMA_TABLE_INFO('CHICAGO_PUBLIC_SCHOOLS_DATA');
```

```
 * sqlite:///FinalDB.db
Done.
```

[12]:
```
[(78,)]
```

[13]:

```
# Retrieving all column names in the CHICAGO_PUBLIC_SCHOOLS_DATA table along␣
↪with their datatypes and length
%sql SELECT name, type, length(type) FROM␣
↪PRAGMA_TABLE_INFO('CHICAGO_PUBLIC_SCHOOLS_DATA');
```

```
* sqlite:///FinalDB.db
Done.
```

[13]:

```
[('School_ID', 'INTEGER', 7),
 ('NAME_OF_SCHOOL', 'TEXT', 4),
 ('Elementary, Middle, or High School', 'TEXT', 4),
 ('Street_Address', 'TEXT', 4),
 ('City', 'TEXT', 4),
 ('State', 'TEXT', 4),
 ('ZIP_Code', 'INTEGER', 7),
 ('Phone_Number', 'TEXT', 4),
 ('Link', 'TEXT', 4),
 ('Network_Manager', 'TEXT', 4),
 ('Collaborative_Name', 'TEXT', 4),
 ('Adequate_Yearly_Progress_Made_', 'TEXT', 4),
 ('Track_Schedule', 'TEXT', 4),
 ('CPS_Performance_Policy_Status', 'TEXT', 4),
 ('CPS_Performance_Policy_Level', 'TEXT', 4),
 ('HEALTHY_SCHOOL_CERTIFIED', 'TEXT', 4),
 ('Safety_Icon', 'TEXT', 4),
 ('SAFETY_SCORE', 'REAL', 4),
 ('Family_Involvement_Icon', 'TEXT', 4),
 ('Family_Involvement_Score', 'TEXT', 4),
 ('Environment_Icon', 'TEXT', 4),
 ('Environment_Score', 'REAL', 4),
 ('Instruction_Icon', 'TEXT', 4),
 ('Instruction_Score', 'REAL', 4),
 ('Leaders_Icon', 'TEXT', 4),
 ('Leaders_Score', 'TEXT', 4),
 ('Teachers_Icon', 'TEXT', 4),
 ('Teachers_Score', 'TEXT', 4),
 ('Parent_Engagement_Icon', 'TEXT', 4),
 ('Parent_Engagement_Score', 'TEXT', 4),
 ('Parent_Environment_Icon', 'TEXT', 4),
 ('Parent_Environment_Score', 'TEXT', 4),
 ('AVERAGE_STUDENT_ATTENDANCE', 'TEXT', 4),
 ('Rate_of_Misconducts__per_100_students_', 'REAL', 4),
 ('Average_Teacher_Attendance', 'TEXT', 4),
 ('Individualized_Education_Program_Compliance_Rate', 'TEXT', 4),
 ('Pk_2_Literacy__', 'TEXT', 4),
 ('Pk_2_Math__', 'TEXT', 4),
 ('Gr3_5_Grade_Level_Math__', 'TEXT', 4),
```

```
 ('Gr3_5_Grade_Level_Read__', 'TEXT', 4),
 ('Gr3_5_Keep_Pace_Read__', 'TEXT', 4),
 ('Gr3_5_Keep_Pace_Math__', 'TEXT', 4),
 ('Gr6_8_Grade_Level_Math__', 'TEXT', 4),
 ('Gr6_8_Grade_Level_Read__', 'TEXT', 4),
 ('Gr6_8_Keep_Pace_Math_', 'TEXT', 4),
 ('Gr6_8_Keep_Pace_Read__', 'TEXT', 4),
 ('Gr_8_Explore_Math__', 'TEXT', 4),
 ('Gr_8_Explore_Read__', 'TEXT', 4),
 ('ISAT_Exceeding_Math__', 'REAL', 4),
 ('ISAT_Exceeding_Reading__', 'REAL', 4),
 ('ISAT_Value_Add_Math', 'REAL', 4),
 ('ISAT_Value_Add_Read', 'REAL', 4),
 ('ISAT_Value_Add_Color_Math', 'TEXT', 4),
 ('ISAT_Value_Add_Color_Read', 'TEXT', 4),
 ('Students_Taking__Algebra__', 'TEXT', 4),
 ('Students_Passing__Algebra__', 'TEXT', 4),
 ('9th Grade EXPLORE (2009)', 'TEXT', 4),
 ('9th Grade EXPLORE (2010)', 'TEXT', 4),
 ('10th Grade PLAN (2009)', 'TEXT', 4),
 ('10th Grade PLAN (2010)', 'TEXT', 4),
 ('Net_Change_EXPLORE_and_PLAN', 'TEXT', 4),
 ('11th Grade Average ACT (2011)', 'TEXT', 4),
 ('Net_Change_PLAN_and_ACT', 'TEXT', 4),
 ('College_Eligibility__', 'TEXT', 4),
 ('Graduation_Rate__', 'TEXT', 4),
 ('College_Enrollment_Rate__', 'TEXT', 4),
 ('COLLEGE_ENROLLMENT', 'INTEGER', 7),
 ('General_Services_Route', 'INTEGER', 7),
 ('Freshman_on_Track_Rate__', 'TEXT', 4),
 ('X_COORDINATE', 'REAL', 4),
 ('Y_COORDINATE', 'REAL', 4),
 ('Latitude', 'REAL', 4),
 ('Longitude', 'REAL', 4),
 ('COMMUNITY_AREA_NUMBER', 'INTEGER', 7),
 ('COMMUNITY_AREA_NAME', 'TEXT', 4),
 ('Ward', 'INTEGER', 7),
 ('Police_District', 'INTEGER', 7),
 ('Location', 'TEXT', 4)]
```

[14]:
```
# Retrieving the number of columns in the CHICAGO_CRIME_DATA table
%sql SELECT COUNT(NAME) FROM PRAGMA_TABLE_INFO('CHICAGO_CRIME_DATA');
```

```
 * sqlite:///FinalDB.db
Done.
```

[14]:
```
[(21,)]
```

```
[15]: # Retrieving all column names in the CHICAGO_CRIME_DATA table along with their
      ↪datatypes and length
      %sql SELECT name, type, length(type) FROM
      ↪PRAGMA_TABLE_INFO('CHICAGO_CRIME_DATA');
```

```
 * sqlite:///FinalDB.db
Done.
```

```
[15]: [('ID', 'INTEGER', 7),
       ('CASE_NUMBER', 'TEXT', 4),
       ('DATE', 'TEXT', 4),
       ('BLOCK', 'TEXT', 4),
       ('IUCR', 'TEXT', 4),
       ('PRIMARY_TYPE', 'TEXT', 4),
       ('DESCRIPTION', 'TEXT', 4),
       ('LOCATION_DESCRIPTION', 'TEXT', 4),
       ('ARREST', 'INTEGER', 7),
       ('DOMESTIC', 'INTEGER', 7),
       ('BEAT', 'INTEGER', 7),
       ('DISTRICT', 'INTEGER', 7),
       ('WARD', 'REAL', 4),
       ('COMMUNITY_AREA_NUMBER', 'REAL', 4),
       ('FBICODE', 'TEXT', 4),
       ('X_COORDINATE', 'REAL', 4),
       ('Y_COORDINATE', 'REAL', 4),
       ('YEAR', 'INTEGER', 7),
       ('LATITUDE', 'REAL', 4),
       ('LONGITUDE', 'REAL', 4),
       ('LOCATION', 'TEXT', 4)]
```

1.2 Problems

Now write and execute SQL queries to solve assignment problems

1.2.1 Problem 1

Find the total number of crimes recorded in the CRIME table.

```
[16]: %sql SELECT COUNT(*) FROM CHICAGO_CRIME_DATA;
```

```
 * sqlite:///FinalDB.db
Done.
```

```
[16]: [(533,)]
```

1.2.2 Problem 2

List community areas with per capita income less than 11000.

[17]:
```
%sql SELECT COMMUNITY_AREA_NAME, PER_CAPITA_INCOME FROM CENSUS_DATA WHERE␣
↪PER_CAPITA_INCOME < 11000;
```

```
 * sqlite:///FinalDB.db
Done.
```

[17]:
```
[('West Garfield Park', 10934),
 ('South Lawndale', 10402),
 ('Fuller Park', 10432),
 ('Riverdale', 8201)]
```

1.2.3 Problem 3

List all case numbers for crimes involving minors? (children are not considered minors for the purposes of crime analysis)

[18]:
```
%%sql
SELECT CASE_NUMBER, DESCRIPTION
FROM CHICAGO_CRIME_DATA
WHERE LOWER(DESCRIPTION) LIKE '%minor%'
ORDER BY CASE_NUMBER;
```

```
 * sqlite:///FinalDB.db
Done.
```

[18]:
```
[('HK238408', 'ILLEGAL CONSUMPTION BY MINOR'),
 ('HL266884', 'SELL/GIVE/DEL LIQUOR TO MINOR')]
```

Version of code that looks for other possibilities of young individuals:

[19]:
```
%%sql
SELECT CASE_NUMBER, DESCRIPTION
FROM CHICAGO_CRIME_DATA
WHERE LOWER(DESCRIPTION) LIKE '%child%'
   OR LOWER(DESCRIPTION) LIKE '%juvenile%'
   OR LOWER(DESCRIPTION) LIKE '%minor%'
   OR LOWER(DESCRIPTION) LIKE '%youth%'
ORDER BY CASE_NUMBER;
```

```
 * sqlite:///FinalDB.db
Done.
```

[19]:
```
[('HK238408', 'ILLEGAL CONSUMPTION BY MINOR'),
 ('HL266884', 'SELL/GIVE/DEL LIQUOR TO MINOR'),
 ('HN144152', 'CHILD ABDUCTION/STRANGER'),
 ('HN567387', 'AGG SEX ASSLT OF CHILD FAM MBR'),
 ('HR391350', 'SEX ASSLT OF CHILD BY FAM MBR')]
```

1.2.4 Problem 4

List all kidnapping crimes involving a child?

[20]:
```
%%sql
SELECT CASE_NUMBER, PRIMARY_TYPE, DESCRIPTION
FROM CHICAGO_CRIME_DATA
WHERE LOWER(DESCRIPTION) LIKE '%child%'
AND LOWER(PRIMARY_TYPE) LIKE '%kidnap%';
```

```
 * sqlite:///FinalDB.db
Done.
```

[20]:
```
[('HN144152', 'KIDNAPPING', 'CHILD ABDUCTION/STRANGER')]
```

1.2.5 Problem 5

What kinds of crimes were recorded at schools?

[21]:
```
%%sql
SELECT DISTINCT(PRIMARY_TYPE)
FROM CHICAGO_CRIME_DATA
WHERE LOWER(LOCATION_DESCRIPTION) LIKE '%school%'
ORDER BY PRIMARY_TYPE;
```

```
 * sqlite:///FinalDB.db
Done.
```

[21]:
```
[('ASSAULT',),
 ('BATTERY',),
 ('CRIMINAL DAMAGE',),
 ('CRIMINAL TRESPASS',),
 ('NARCOTICS',),
 ('PUBLIC PEACE VIOLATION',)]
```

1.2.6 Problem 6

List the average safety score for each type of school.

[22]:
```
%%sql
SELECT "Elementary, Middle, or High School", AVG(SAFETY_SCORE)
FROM CHICAGO_PUBLIC_SCHOOLS_DATA
GROUP BY "Elementary, Middle, or High School"
ORDER BY "Elementary, Middle, or High School";
```

```
 * sqlite:///FinalDB.db
Done.
```

[22]:
```
[('ES', 49.52038369304557), ('HS', 49.62352941176471), ('MS', 48.0)]
```

1.2.7 Problem 7

List 5 community areas with highest % of households below poverty line

[23]:
```
%%sql
SELECT COMMUNITY_AREA_NAME, PERCENT_HOUSEHOLDS_BELOW_POVERTY
FROM CENSUS_DATA
ORDER BY PERCENT_HOUSEHOLDS_BELOW_POVERTY DESC
LIMIT 5;
```

```
 * sqlite:///FinalDB.db
Done.
```

[23]:
```
[('Riverdale', 56.5),
 ('Fuller Park', 51.2),
 ('Englewood', 46.6),
 ('North Lawndale', 43.1),
 ('East Garfield Park', 42.4)]
```

1.2.8 Problem 8

Which community area is most crime prone?

[24]:
```
%%sql
SELECT COMMUNITY_AREA_NUMBER, COMMUNITY_AREA_NAME
FROM CENSUS_DATA
WHERE COMMUNITY_AREA_NUMBER IN
(SELECT COMMUNITY_AREA_NUMBER
 FROM CHICAGO_CRIME_DATA
 GROUP BY COMMUNITY_AREA_NUMBER
 ORDER BY COUNT(COMMUNITY_AREA_NUMBER) DESC
 LIMIT 1);
```

```
 * sqlite:///FinalDB.db
Done.
```

[24]:
```
[(25.0, 'Austin')]
```

Double-click **here** for a hint

1.2.9 Problem 9

Use a sub-query to find the name of the community area with highest hardship index

[25]:
```
%%sql
SELECT COMMUNITY_AREA_NAME
FROM CENSUS_DATA
WHERE COMMUNITY_AREA_NUMBER IN
(SELECT COMMUNITY_AREA_NUMBER
 FROM CENSUS_DATA
 WHERE HARDSHIP_INDEX != 'None'
```

```
ORDER BY HARDSHIP_INDEX DESC
LIMIT 1);
```

```
 * sqlite:///FinalDB.db
Done.
```

[25]: [('Riverdale',)]

[26]:
```
%%sql
SELECT COMMUNITY_AREA_NAME
FROM CENSUS_DATA
WHERE HARDSHIP_INDEX = (
    SELECT MAX(HARDSHIP_INDEX)
    FROM CENSUS_DATA);
```

```
 * sqlite:///FinalDB.db
Done.
```

[26]: [('Riverdale',)]

1.2.10 Problem 10

Use a sub-query to determine the Community Area Name with most number of crimes?

[27]:
```
%%sql
SELECT COMMUNITY_AREA_NAME
FROM CENSUS_DATA
WHERE COMMUNITY_AREA_NUMBER IN
(SELECT COMMUNITY_AREA_NUMBER
 FROM CHICAGO_CRIME_DATA
 GROUP BY COMMUNITY_AREA_NUMBER
 ORDER BY COUNT(COMMUNITY_AREA_NUMBER) DESC
 LIMIT 1);
```

```
 * sqlite:///FinalDB.db
Done.
```

[27]: [('Austin',)]

1.3 Author(s)

Hima Vasudevan

Rav Ahuja

Ramesh Sannreddy

1.4 Contribtuor(s)

Malika Singla

1.5 Change log

Date	Version	Changed by	Change Description
2022-03-04	2.5	Lakshmi Holla	Changed markdown.
2021-05-19	2.4	Lakshmi Holla	Updated the question
2021-04-30	2.3	Malika Singla	Updated the libraries
2021-01-15	2.2	Rav Ahuja	Removed problem 11 and fixed changelog
2020-11-25	2.1	Ramesh Sannareddy	Updated the problem statements, and datasets
2020-09-05	2.0	Malika Singla	Moved lab to course repo in GitLab
2018-07-18	1.0	Rav Ahuja	Several updates including loading instructions
2018-05-04	0.1	Hima Vasudevan	Created initial version

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